

Module 1: Session-3

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ANSWERS:

TASK 1:

HTTP (Hypertext Transfer Protocol) is the standard protocol used to transfer web pages and other data over the internet. HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP that encrypts data to protect communication between users and websites. FTP (File Transfer Protocol) is used to transfer files between computers over a network. SMTP (Simple Mail Transfer Protocol) is the protocol used for sending emails between mail servers and email clients.

TASK 2:

IPv4 uses 32-bit addresses and supports many unique addresses (ex: 192.168.1.1)

IPv6 uses 128-bit addresses and supports a vastly large number of unique addresses (ex: 2001:0db8:85a3::8a2e:0370:7334)

TASK 3:

142.250.180.14

TASK 4:

The HTTP status code 200 means the request was successful and the server returned the requested resource. 301 means the resource has been permanently moved to a new location. 403 indicates that access to the requested resource is forbidden. 404 means the requested page or resource could not be found, and 500 indicates that the server encountered an internal error while processing the request.

TASK 5:

The DNS hierarchy helps computers find websites on the internet. In www.google.com com is the Top-Level Domain (TLD), google is the domain name, and www is the subdomain. The root (.) is the highest level of the DNS system and is usually not shown. Together, these parts help direct users to the correct website.